

ELECTRIC CIRCUITS

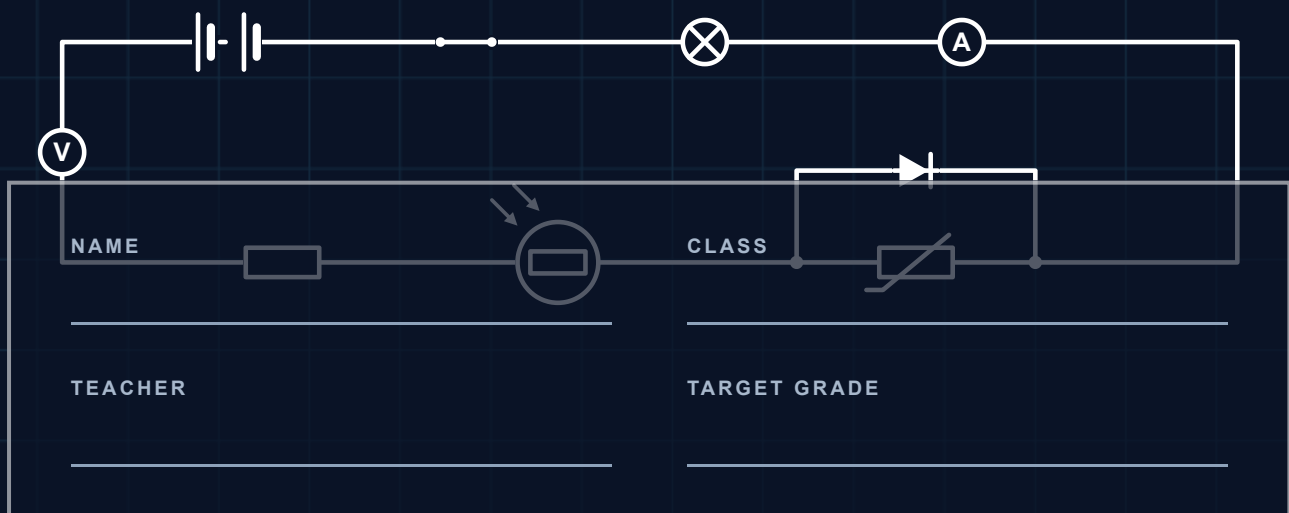
PART 1 CHARGE, ENERGY AND RESISTANCE

LESSONS 1–4

From your first circuit to $V = IR$: circuits and symbols, current, potential difference and resistance. Learn with it in class and revise from it at home.

1. Complete circuits and symbols
2. Current and charge
3. Potential difference
4. Resistance and $V = IR$

ANSWER EDITION · FOR TEACHERS AND SELF-MARKING



HOW THIS BOOKLET WORKS

This booklet assumes you have **never studied electricity before**. Every lesson starts from the ideas you already have and builds one step at a time. Keep it: it is your classwork book *and* your revision guide.

 DO NOW Quick questions from earlier lessons. Remembering again is what makes it stick.	 LEARN IT The explanation, diagrams and worked examples. Read this again when you revise.	 TRY IT Practice that builds up: first with support, then on your own.
 LAB Record measurements from a virtual lab or a required practical.	 EXAM CORNER An AQA-style question with a tip on how the marks are given.	 REVISE IT Summary, cover-and-answer questions and an “I can” check to colour in.

 **Revising at home?** For each lesson: (1) read the Learn it pages, (2) cover them and answer the “Cover the page” questions in Revise it, (3) check with the quick answers at the back, (4) do the matching mission on the companion website, which has more questions with answers.

THE ROUTE THROUGH THE UNIT · THIS PART IN BOLD

#	LESSON	COMPANION WEBSITE
01	Complete circuits and symbols What a circuit needs, circuit symbols and drawing rules, measuring current	Mission 1 · Build & measure
02	Current and charge Current as a rate of flow of charge, $Q = I t$, current is not used up	Mission 2 · Charge on the move
03	Potential difference p.d. as energy per coulomb, voltmeters, $E = Q V$, p.d.s add in series	Mission 3 · Energy per coulomb
04	Resistance and $V = I R$ Resistance, $V = I R$, ohmic conductors, why heating raises resistance	Mission 4 · Resistance & $V = I R$
05	Resistance of a wire REQUIRED PRACTICAL Required practical: how the length of a wire affects its resistance	Required practical · Resistance
06	Series and parallel circuits Current, p.d. and resistance rules for series and parallel circuits	Mission 5 · Series & parallel
07	Resistors in series and parallel REQUIRED PRACTICAL Required practical: measure total resistance of two resistors in series and in parallel	Required practical · Resistance (Part B)
08	I–V graphs I–V graphs of a fixed resistor, filament lamp and diode	Mission 6 · I–V characteristics
09	Investigating I–V characteristics REQUIRED PRACTICAL Required practical: measure the I–V characteristics of a resistor, lamp and diode	Required practical · I–V characteristics
10	Thermistors, LDRs and sensors How thermistors and LDRs respond to temperature and light, and sensor circuits	Mission 6 · Thermistor and LDR
11	Power and energy transfer Power, $E = P t$, $P = V I$, $P = I^2 R$, energy transfers in appliances	Mission 7 · Power & the National Grid
12	Mains, safety and the National Grid ac and dc, 230 V 50 Hz mains, three-core cable, earthing, the National Grid	Missions 7 and 8

Companion website: panphy.app/gcsephy/year10phy/unit01 · missions, interactive equation triangles, required-practical pages and the Exam Zone.

EQUATION TOOLKIT

The equations you have met so far. You will meet these one at a time. Learn each equation **in words** as well as symbols: your exams may or may not give you an equation sheet, and knowing them by heart makes every question faster.

$$Q = I t$$

charge flow = current $\times$ time

C, A, s

Rearranged: $I = Q \div t$ $t = Q \div I$

Lesson 2

$$E = Q V$$

energy transferred = charge $\times$ p.d.

J, C, V

Rearranged: $V = E \div Q$ $Q = E \div V$

Lesson 3

$$V = I R$$

p.d. = current $\times$ resistance

V, A, Ω

Rearranged: $I = V \div R$ $R = V \div I$

Lesson 4

FOUR STEPS FOR EVERY CALCULATION

1

Write down what you know, with units. Convert units first (minutes $\rightarrow$ seconds, kW $\rightarrow$ W, mA $\rightarrow$ A).

2

Write the equation you will use, in symbols.

3

Substitute the numbers, then rearrange if needed.

4

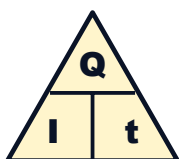
Give the answer with a unit. Ask: is it sensible?

CONVERT	HOW	EXAMPLE
minutes $\rightarrow$ seconds	$\times 60$	3.0 min = 180 s
hours $\rightarrow$ seconds	$\times 3600$	2.0 h = 7200 s
milliamps (mA) $\rightarrow$ amps (A)	$\div 1000$	250 mA = 0.25 A
kilowatts (kW) $\rightarrow$ watts (W) $\cdot$ kilohms (k Ω) $\rightarrow$ ohms (Ω)	$\times 1000$	2.2 kW = 2200 W
kilovolts (kV) $\rightarrow$ volts (V)	$\times 1000$	400 kV = 400 000 V

EQUATION TRIANGLES: HELP WITH REARRANGING

- 1 Cover the quantity you want to find.
- 2 The two left **side by side**? **Multiply** them.
- 3 One **above** the other? **Divide** the top by the bottom.

Example: to find I from $Q = I t$, cover I. Q is above t, so $I = Q \div t$. Triangles only help with rearranging: you still need to learn each equation.

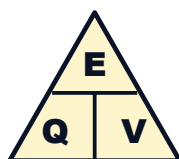


$$Q = I t$$

$$Q = I \times t$$

$$I = Q \div t$$

$$t = Q \div I$$

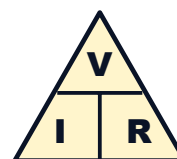


$$E = Q V$$

$$E = Q \times V$$

$$Q = E \div V$$

$$V = E \div Q$$



$$V = I R$$

$$V = I \times R$$

$$I = V \div R$$

$$R = V \div I$$

Interactive version: panphy.app/gcsephy/year10phy/unit01/equation-triangles.

COMPLETE CIRCUITS AND SYMBOLS

AQA Combined 6.2.1.1 · Physics 4.2.1.1

BIG QUESTION Your torch won't switch on. The bulb is fine and the batteries are new. What could be wrong?

BY THE END YOU CAN

- explain why a lamp only lights in a complete loop
- recognise and draw standard circuit symbols
- turn a real circuit into a neat circuit diagram
- connect an ammeter in series to measure current

circuit

component

cell

battery

switch

circuit diagram

series

ammeter

ONLINE COMPANION

Mission 1 · Build & measure

Notes on circuits, symbols and drawing rules, with 5 practice questions and worked answers.

panphy.app/gcsephy/year10phy/unit01/lesson/build-and-measure/



RETRIEVE

WHAT DO YOU ALREADY THINK?

No wrong answers: this is your starting point.

1

Name two things that run on a battery and two that plug into the wall.

e.g. torch, phone; kettle, TV.

2

When a lamp is on, what do you think is moving in the wires?

Tiny charged particles

(electrons): you learn this today.

3

Why does a torch need a switch?

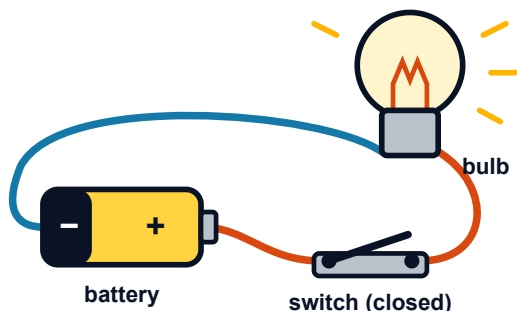
To break or complete the circuit, turning it off and on.



EXPLAIN

CIRCUITS FROM SCRATCH

WHAT DOES A CIRCUIT NEED?



A real circuit: battery, switch and bulb joined into one complete loop.

A **circuit** is a complete loop that electric charge can flow around. To light a lamp you need:

1. a **source of energy**: a cell or battery pushes the charge around;
2. a **complete loop** of wire with no gaps;
3. **components**: things that do a job, such as a lamp.

THINK OF IT LIKE A BIKE CHAIN

The chain is already full of links, just as wires are already full of charge. When you push the pedals (the cell), the *whole* chain moves at once, and the back wheel (the lamp) turns immediately. Break the chain anywhere and everything stops.

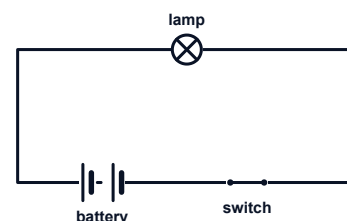
Where the model breaks: you can see a chain move quickly, but the charges in a wire drift very slowly.

! No complete loop, no current. A gap anywhere (an open switch, a loose wire, a broken bulb) stops the flow of charge *everywhere* in the loop.

FROM A PICTURE TO A CIRCUIT DIAGRAM

Drawing real objects takes a long time, and everyone draws them differently. Scientists and electricians everywhere use the same **circuit symbols**, so anyone can read a **circuit diagram**. This diagram shows exactly the same circuit as the picture above.

Learn the first row of symbols now. The dashed ones come later in the unit: you do not need them yet.



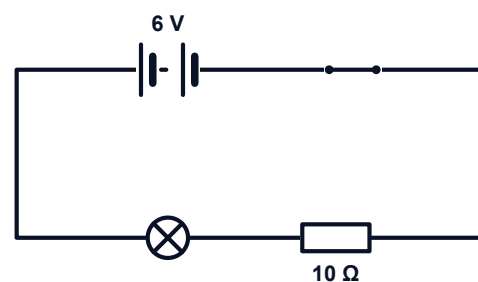
The same circuit as a circuit diagram.

Cell long line is +	Battery two or more cells	Switch (open) gap: no current	Switch (closed) loop complete	Lamp gives out light	Resistor limits the current	Ammeter measures current
Voltmeter Lesson 3	Variable resistor Lesson 4	Diode Lesson 8	LED Lesson 8	Thermistor Lesson 10	LDR Lesson 10	Fuse Lesson 12

Solid boxes: learn today. Dashed boxes: the lesson where you meet them is shown.

RULES FOR DRAWING CIRCUIT DIAGRAMS

- 1 Use a **pencil and a ruler**.
- 2 Draw wires as **straight lines** with **right-angle corners**.
- 3 Leave **no gaps**: every wire joins something at both ends.
- 4 Put each symbol **in** the wire, so the wire goes into it on both sides.
- 5 Write values such as 6 V or 10 Ω next to the symbol.



A neat diagram: ruler, right angles, no gaps.

MEASURING THE CURRENT

The flow of charge is called the **current**. We measure it in **amperes (A)**, or 'amps', with an **ammeter**. The ammeter goes *in* the loop so that the charge flows through it. This is called connecting it **in series**.

! A circuit with just one loop is a series circuit. Everything in that loop is 'in series'.

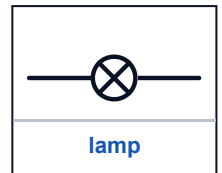
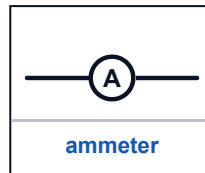
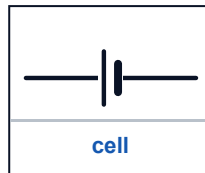
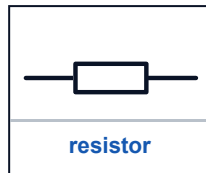
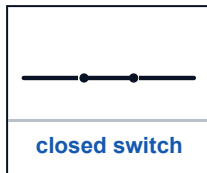
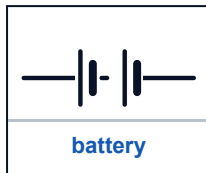


The ammeter is in series with the lamp.

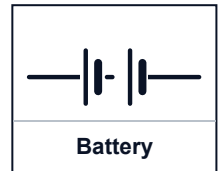
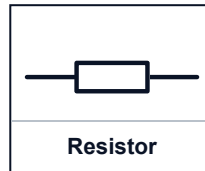
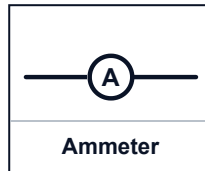
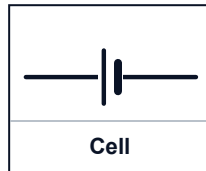
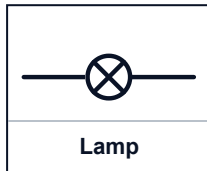
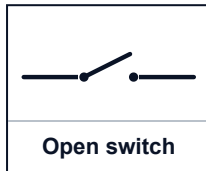


PRACTISE
TRY IT

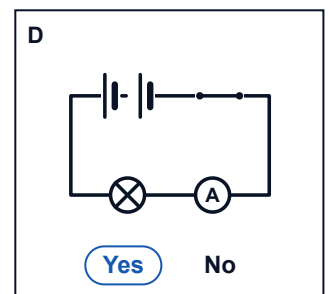
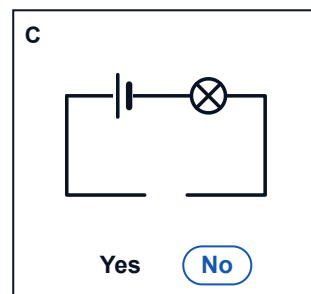
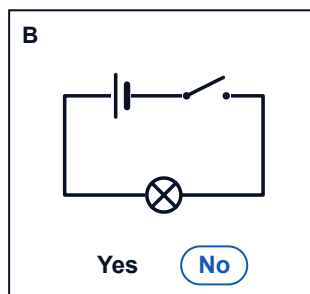
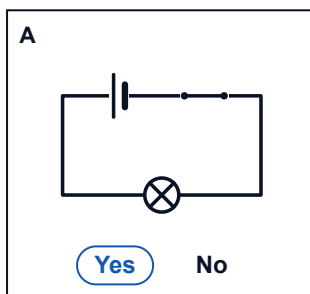
1 Name each symbol.



2 Draw the symbol for each component. Use a ruler for the straight lines.



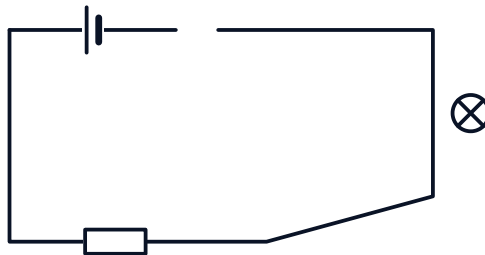
3 Will the lamp light? Circle **Yes** or **No** for each circuit.



Explain your answer for circuit C.

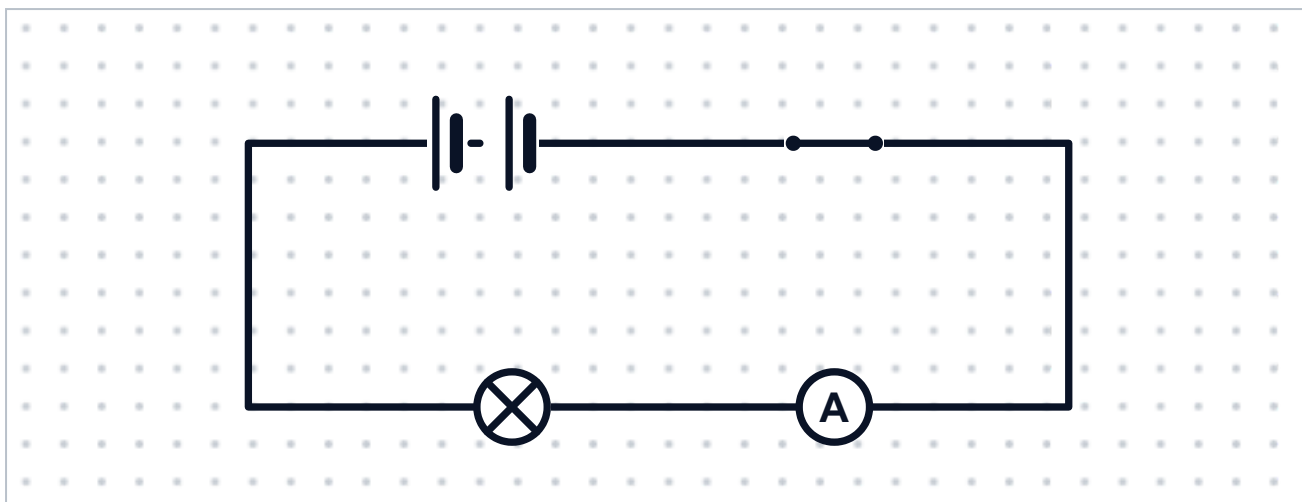
C has a gap in the bottom wire, so the loop is not complete and no charge can flow.

4 Jo drew this circuit diagram. Find **three** mistakes.



1. There is a gap in the top wire. 2. One wire is slanted: wires should be straight with right-angle corners. 3. The lamp is not connected into the wire (the wire does not go into it).

- 5 Draw a circuit diagram for a **battery**, a **closed switch**, a **lamp** and an **ammeter**, all in one loop.



- 6 **Torch detective.** Look back at the big question. Suggest **two** faults that could stop the torch working, and how you could test for each one.

e.g. a loose or broken wire/connection, leaving a gap: check or replace each connection. The switch does not close properly: replace it with a working switch. The batteries are in the wrong way round: turn them round.



AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Examiners look for correct symbols, a complete loop with no gaps, and the ammeter in series.

- 7 Draw a circuit diagram to show a battery, a resistor and a lamp connected in series, with an ammeter to measure the current. **[3 marks]**

Correct symbols for battery, resistor and lamp (1); all in one complete loop with no gaps (1); ammeter in series in the same loop (1).

- 8 State what is meant by a *series* circuit. **[1 mark]**

The components are connected one after another in a single loop.



SUMMARY

- A circuit needs a source (cell or battery) and a **complete loop**.
- A gap anywhere stops the current everywhere.
- Circuit diagrams use standard symbols, drawn with a ruler: straight wires, right-angle corners, no gaps.
- An ammeter measures current and goes **in series**.

COVER THE PAGE AND ANSWER

1 What two things does a circuit need for a current to flow?
.....

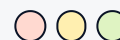
2 What is the difference between a cell and a battery?
.....

3 What does an open switch do?
.....

4 How is an ammeter connected?
.....

I CAN...

explain why a circuit must be a complete loop



recognise and draw the symbols for a cell, battery, switch, lamp, resistor and ammeter



draw a neat circuit diagram using the rules



connect an ammeter in series



CURRENT AND CHARGE

AQA Combined 6.2.1.2 · Physics 4.2.1.2 · Virtual Lab 1

BIG QUESTION Flick a switch and the lamp lights instantly, yet each electron drifts through the wire slower than a snail. How can both be true?

BY THE END YOU CAN

- describe current as the rate of flow of charge
- use $Q = It$, with time in seconds
- explain why the current is the same everywhere in a single loop

charge (C)

current (A)

coulomb

ampere

electron

conventional current

ONLINE COMPANION

Mission 2 · Charge on the move

Revision notes and 5 practice questions with worked answers.

panphy.app/gcsephy/year10phy/unit01/lesson/charge-and-current/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

Draw the symbols for a cell and a lamp.

Cell: long thin line and short

thick line. Lamp: circle with a

cross.

2

How must an ammeter be connected?

In series (in the loop).

3

Give two reasons why a lamp in a circuit might not light.

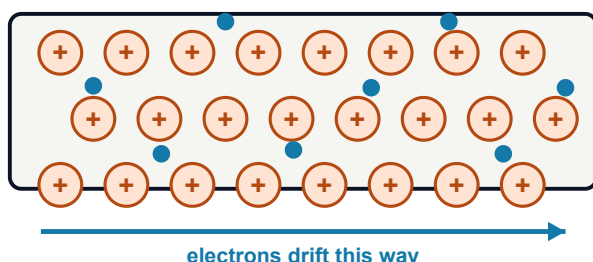
Open switch; gap or loose wire;

broken bulb; no cell.



EXPLAIN

WHAT IS FLOWING?



Inside a metal wire: fixed positive ions (orange) and free electrons (blue).

Metals contain **free electrons**: tiny particles with a negative charge that can move between the atoms. The wires are full of them *before* you switch on.

The cell pushes on all of them at once, so the whole loop starts moving together, just like the bike chain. The lamp lights straight away, even though each electron only drifts a few millimetres per second.

Charge is measured in **coulombs (C)**. One electron has a tiny charge: it takes about 6×10^{18} (six billion billion) electrons to make 1 C.

■ CURRENT IS A RATE

Current is the **rate of flow of charge**: how many coulombs pass a point *each second*.

THINK OF A TICKET BARRIER

Count the people going through a station barrier each second. The current is like the number of people *per second*, not the number of people in the station.

Where the model breaks: people can stop or change direction on their own; in a single loop all the charges move together.

! 1 ampere = 1 coulomb per second. A current of 0.5 A means 0.5 C of charge passes a point every second.

$$Q = I t$$

charge flow = current $\times$ time

Q in coulombs (C) $\cdot$ I in amperes (A) $\cdot$ t in seconds (s)

Rearranged: $I = Q \div t$ and $t = Q \div I$



WORKED EXAMPLE

A current of 0.40 A flows for 3.0 minutes. Calculate the charge that flows.

KNOW $I = 0.40 \text{ A}$, $t = 3.0 \times 60 = 180 \text{ s}$

EQUATION $Q = I t$

SUBSTITUTE $Q = 0.40 \times 180$

ANSWER $Q = 72 \text{ C}$

YOUR TURN

A current of 0.50 A flows for 4.0 minutes. Calculate the charge that flows.

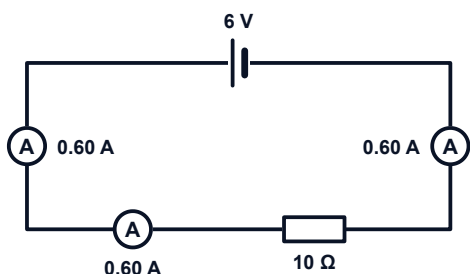
KNOW $I = 0.50 \text{ A}$, $T = 240 \text{ S}$

EQUATION $Q = I T$

SUBSTITUTE $Q = 0.50 \times 240$

ANSWER $Q = 120 \text{ C}$

CURRENT IS NOT USED UP



Three ammeters in one loop give the same reading.

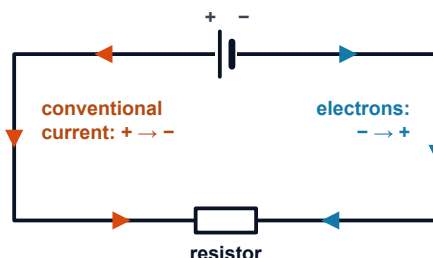
In a single loop, the current is **the same at every point**. Charge is not destroyed in a component: it carries on round the circuit. What the component takes is **energy** (Lesson 3).

! Charge goes round. Energy is transferred.

WHICH WAY DOES CURRENT FLOW?

Outside the cell, electrons flow from the **negative (-)** terminal to the **positive (+)** terminal.

But circuit diagrams show **conventional current**, from + to -. This was agreed before electrons were discovered. Both describe the same current, and exam diagrams use conventional current.



Orange: conventional current (+ to -). Blue: electron flow (- to +). Same current, opposite arrows.



PRACTISE
TRY IT

- 1 Complete the sentences using the word bank.

Current is the rate of flow of charge. It is measured in amperes (A) using an ammeter connected in series. Charge is measured in coulombs (C). A current of 2 A means 2 coulombs pass a point every second.

WORD BANK

charge

series

ammeter

second

amperes (A)

coulombs (C)

2

parallel

volts

- 2 A current of 0.60 A flows in a circuit. Which statement is correct? Tick **one** box.

☐

There is 0.60 C of charge in the circuit.

☒

0.60 C of charge passes any point each second.

☐

The lamp uses up 0.60 C each second.

☐

The current is 0.60 A only near the cell.

- 3 **SUPPORTED** A current of 2.0 A flows for 30 s. Calculate the charge that flows.

EQUATION	$Q = I t$	
SUBSTITUTE	$Q = 2.0 \times 30$	
ANSWER	60 C	unit:

- 4 **SUPPORTED** A current of 0.30 A flows for 5.0 minutes. Calculate the charge. *Convert the time first!*

CONVERT	$t = 5.0 \times 60 = 300 \text{ s}$	
EQUATION	$Q = I t$	
SUBSTITUTE	$Q = 0.30 \times 300$	
ANSWER	90 C	unit:

- 5 **ON YOUR OWN** 36 C of charge passes through a lamp in 12 s. Calculate the current.

$$I = Q \div t = 36 \div 12 = 3.0 \text{ A}$$

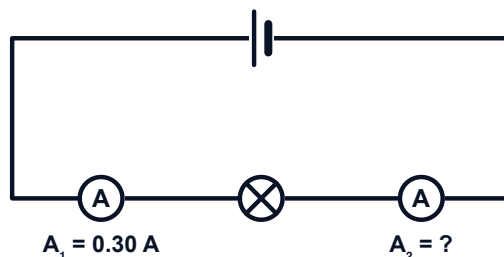
- 6 **CHALLENGE** A phone charger supplies 0.80 A. How long does it take for 240 C to flow? Give your answer in seconds and in minutes.

$$t = Q \div I = 240 \div 0.80 = 300 \text{ s} = 5.0 \text{ minutes}$$

7 Predict the reading on A_2 . Explain why Mia is wrong.

Mia

The lamp uses up some current, so ammeter A_2 after the lamp will read less than A_1 .



$A_2 = 0.30 \text{ A}$. It is a single loop, so the current is the same everywhere. Charge is not used up: the lamp transfers energy, not charge.



INVESTIGATE

VIRTUAL LAB 1: IS CURRENT USED UP?

PhET Circuit Construction Kit: DC → Lab. Full steps:
Virtual Labs worksheet pp. 3–6.

Build one loop with a **6 V battery** and a **10 Ω resistor**. Use the non-contact ammeter to measure the current at three places.

8 Record your readings.

Position	Current / A	Current with 20 Ω / A
Before the resistor	0.60	0.30
After the resistor	0.60	0.30
Next to the battery	0.60	0.30

9 What do your readings show? Use numbers in your answer.

All three readings are the same (0.60 A), so current is not used up. With 20 Ω the current halves (0.30 A) but is still the same everywhere.



Get the marks: Convert minutes to seconds before you substitute. The unit (C) is part of the answer.

- 10** A motor draws a current of 0.60 A for 5.0 minutes. Calculate the charge that flows through the motor. **[3 marks]**

EQUATION	$Q = I t$, with $t = 5.0 \times 60 = 300 \text{ s}$	
SUBSTITUTE	$Q = 0.60 \times 300$	
ANSWER	180 C	unit:

- 11** A student says the current leaving the motor is smaller than the current entering it. Explain why the student is wrong. **[2 marks]**

The motor is in a single loop, so the current is the same everywhere (1). Charge is not used up; the motor transfers energy (1).

**SUMMARY**

- Current is the **rate of flow of charge**. In metal wires the charges are free electrons.
- Charge: coulombs (C). Current: amperes (A). $1 \text{ A} = 1 \text{ C per second}$.
- $Q = I t$, with t in seconds.
- In a single loop, current is the same everywhere: charge is not used up.
- Conventional current: + to -. Electrons: - to +.

COVER THE PAGE AND ANSWER

1 What is current?

2 What is the unit of charge?

3 Write the equation linking charge, current and time.

4 0.2 A flows into a lamp. What current flows out?

I CAN...

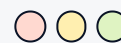
describe current as the rate of flow of charge



calculate charge, current or time using $Q = I t$



convert minutes to seconds before calculating



explain why current is not used up in a loop



POTENTIAL DIFFERENCE

AQA Combined 6.2.1.3, 6.2.4.2 · Physics 4.2.1.3, 4.2.4.2 · Virtual Lab 2

BIG QUESTION An AA battery says 1.5 V. A car battery says 12 V. What is that number actually telling you?

BY THE END YOU CAN

- explain p.d. as the energy transferred per coulomb
- connect a voltmeter in parallel across a component
- use $E = QV$
- explain why the p.d.s in a loop add up to the supply p.d.

potential difference (p.d.)

volt (V)

voltmeter

parallel

energy

joule (J)

ONLINE COMPANION

Mission 3 · Energy per coulomb

Revision notes and 5 practice questions with worked answers.

panphy.app/gcsephy/year10phy/unit01/lesson/potential-difference/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

What is current?

The rate of flow of charge.

2

0.50 A flows for 20 s. How much charge flows?

$Q = 0.50 \times 20 = 10 \text{ C}$

3

Does a lamp use up current? Explain.

No. The current is the same all round the loop; the lamp transfers energy.



EXPLAIN

ENERGY CARRIED BY CHARGE

The cell does not make charge: the charge was already in the wires. The cell gives **energy** to the charge. When the charge passes through a component such as a lamp, energy is transferred to it, so the lamp lights up and gets hot.

THINK OF DELIVERY VANS

Each coulomb of charge is a delivery van on a circular route. The battery loads each van with energy. At the lamp the van drops its energy off, then drives on, empty, back to the battery to reload. The vans (charge) are never used up; only the parcels (energy) are delivered.

Where the model breaks: real charges do not carry parcels, and energy reaches the lamp as soon as the switch closes, not after a slow journey.

WHAT A VOLT MEANS

Potential difference (p.d.) is the **energy transferred per coulomb** of charge between two points. It is measured in **volts (V)**. People often call it 'voltage'.



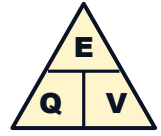
1 volt = 1 joule per coulomb. A p.d. of 6 V across a lamp means 6 J of energy is transferred to the lamp by every 1 C of charge that passes through it.

energy transferred = charge × potential difference

$$E = QV$$

E in joules (J) · Q in coulombs (C) · V in volts (V)

Rearranged: $V = E \div Q$ and $Q = E \div V$



WORKED EXAMPLE

6.0 C of charge passes through a lamp with 12 V across it. Calculate the energy transferred.

KNOW $Q = 6.0 \text{ C}$, $V = 12 \text{ V}$

EQUATION $E = QV$

SUBSTITUTE $E = 6.0 \times 12$

ANSWER **$E = 72 \text{ J}$**

YOUR TURN

A heater transfers 360 J while 30 C passes through it. Calculate the p.d. across it.

KNOW $E = 360 \text{ J}$, $Q = 30 \text{ C}$

EQUATION $V = E \div Q$

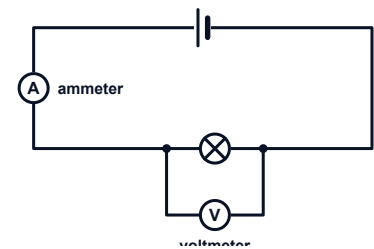
SUBSTITUTE $V = 360 \div 30$

ANSWER **$V = 12 \text{ V}$**

MEASURING P.D.: THE VOLTMETER

A p.d. is always *between two points*, so a **voltmeter** is connected **across** a component, with one lead on each side. This is connecting it **in parallel**.

Say 'the p.d. **across** the lamp' and 'the current **through** the lamp'.



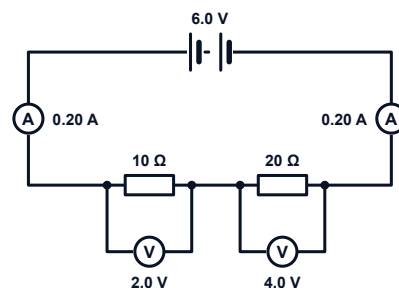
Ammeter in series (in the loop). Voltmeter in parallel (across the lamp).

	CURRENT	POTENTIAL DIFFERENCE
What it is	rate of flow of charge (coulombs per second)	energy transferred per coulomb (joules per coulomb)
Unit	ampere, A	volt, V
Measured with	ammeter, in series	voltmeter, in parallel
Say	current <i>through</i>	p.d. <i>across</i>

ENERGY IN = ENERGY OUT

In a single loop, the p.d.s across the components **add up to the p.d. of the supply**.

Here each coulomb gets 6 J from the battery and hands over all 6 J on the way round: 2 J in the 10 Ω resistor and 4 J in the 20 Ω resistor. Energy is conserved.

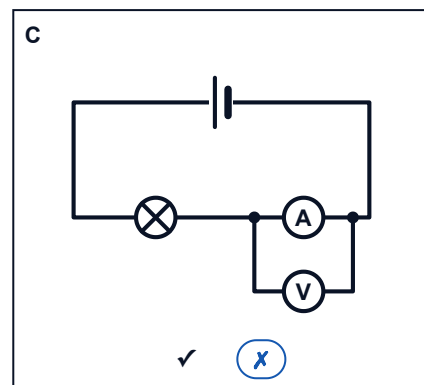
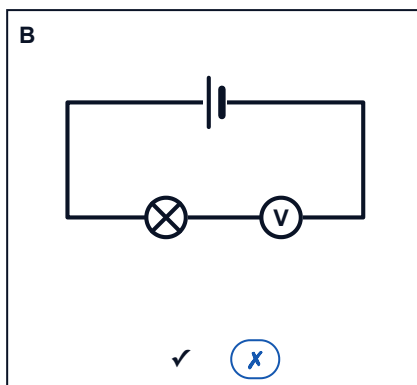
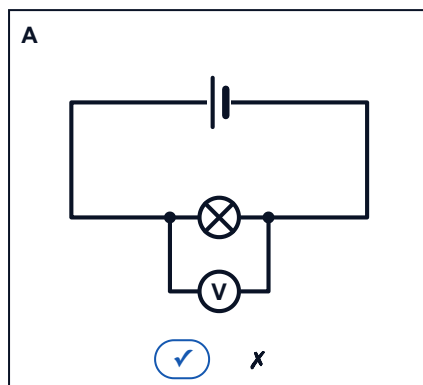


Series: the same current everywhere; $2.0 \text{ V} + 4.0 \text{ V} = 6.0 \text{ V}$.



PRACTISE
TRY IT

- 1 Which circuit measures the p.d. across the lamp correctly? Circle ✓ or ✗ for each.



A only. In B the voltmeter is in series; in C it is across the ammeter, not the lamp.

- 2 Complete the sentences.

A p.d. of 9 V across a resistor means 9 J of energy is transferred to the resistor for every coulomb of charge. A voltmeter is connected across a component, in parallel.

WORD BANK 9 coulomb second across parallel series

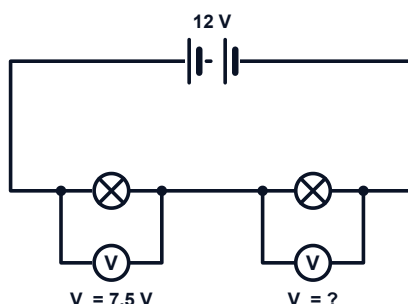
- 3 **SUPPORTED** 25 C of charge passes through a resistor with 4.0 V across it. Calculate the energy transferred.

EQUATION	$E = Q V$
SUBSTITUTE	$E = 25 \times 4.0$
ANSWER	100 J unit:

- 4 **ON YOUR OWN** A kettle element transfers 23 000 J of energy when 100 C of charge flows through it. Calculate the p.d. across the element.

$$V = E \div Q = 23\,000 \div 100 = 230\text{ V}$$

- 5 (a) What does voltmeter V_2 read? (b) How much energy does each coulomb transfer in the second lamp?



(a) $12 - 7.5 = 4.5\text{ V}$. (b) 4.5 J per coulomb.

6 Explain what is wrong with Leo's idea.

Leo

A 9 V battery pushes the charges faster than a 1.5 V battery. That's what volts measure.

Volts measure energy transferred per coulomb, not speed. A 9 V battery gives each coulomb 9 J of energy; a 1.5 V battery gives each coulomb only 1.5 J.



INVESTIGATE

VIRTUAL LAB 2: WHAT DOES A VOLT MEAN?

Full steps: Virtual Labs worksheet pp. 7–11.

Use one $10\ \Omega$ resistor. At each battery setting, measure the p.d. across the resistor. Then calculate the energy transferred when 2 C passes.

7 Record and calculate.

Battery setting / V	p.d. across resistor / V	Energy for 2 C / J
3	3.0	6.0
6	6.0	12
9	9.0	18

8 Add a second $10\ \Omega$ resistor in series (6 V battery). Measure the p.d. across each resistor. What do you notice?

About 3.0 V across each; $3.0\ \text{V} + 3.0\ \text{V} = 6.0\ \text{V}$, the battery p.d. The energy from the battery is shared between the resistors.



AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Explaining a p.d. needs two ideas: energy (joules) and per coulomb of charge.

9 A 9.0 V battery moves 20 C of charge around a circuit. Calculate the energy transferred by the battery. [2 marks]

EQUATION	$E = QV$
SUBSTITUTE	$E = 20 \times 9.0$
ANSWER	180 J unit:

10 A voltmeter across one resistor in this circuit reads 3.0 V. Explain what this reading means. [2 marks]

3.0 J of energy is transferred (1) to the resistor for every coulomb of charge passing through it (1).



SUMMARY

- p.d. is the **energy transferred per coulomb**: $1\text{ V} = 1\text{ J/C}$.
- A voltmeter goes **across** a component (in parallel).
- **$E = QV$**
- Charge is not used up; energy is transferred.
- In a series loop, the component p.d.s add up to the supply p.d.

COVER THE PAGE AND ANSWER

1 What is potential difference?

3 Write the equation linking energy, charge and p.d.

2 How is a voltmeter connected?

4 A 6 V battery; one of two series lamps has 2 V across it. What does the other have?

I CAN...

explain p.d. as energy transferred per coulomb



connect a voltmeter in parallel



use $E = QV$ and its rearrangements



use 'p.d.s add up' in a series loop



RESISTANCE AND $V = IR$

AQA Combined 6.2.1.3–6.2.1.4 · Physics 4.2.1.3–4.2.1.4 · Virtual Lab 3

BIG QUESTION A dimmer switch makes a lamp glow less brightly. What is it actually changing?

BY THE END YOU CAN

- describe resistance and its unit, the ohm (Ω)
- use $V = IR$, including mA and k Ω
- recognise an ohmic conductor from data or a graph
- explain why heating a metal increases its resistance

resistance

ohm (Ω)

ohmic conductor

directly proportional

variable resistor

ONLINE COMPANION

Mission 4 · Resistance & $V = IR$

Revision notes and 5 practice questions with worked answers.

panphy.app/gcsephy/year10phy/unit01/lesson/resistance/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

What does '6 V across a lamp' mean?

6 J of energy is transferred per coulomb.

2

Which meter measures p.d., and how is it connected?

Voltmeter, in parallel (across).

3

Two lamps are in series with a 6 V battery. One has 2 V across it. What is across the other?

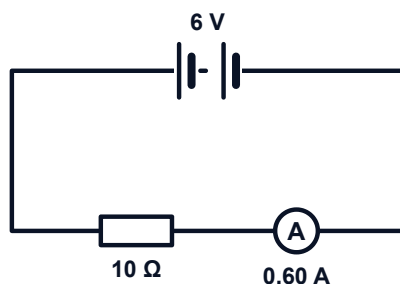
4 V



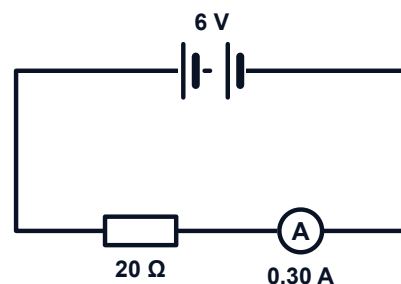
EXPLAIN

CONTROLLING THE CURRENT

Resistance is how much a component **opposes the current**. For the same p.d., a bigger resistance means a smaller current. Resistance is measured in **ohms (Ω)**. A **resistor** has a chosen resistance, such as 10 Ω , to control the current. A **variable resistor** (the dimmer) lets you change it.



Smaller resistance, bigger current.



Double the resistance, half the current.

$$V = IR$$

potential difference = current $\times$ resistance

V in volts (V) · I in amperes (A) · R in ohms (Ω)

Rearranged: $I = V \div R$ and $R = V \div I$



WORKED EXAMPLE

A resistor has 6.0 V across it and a current of 0.30 A through it. Calculate its resistance.

KNOW $V = 6.0 \text{ V}$, $I = 0.30 \text{ A}$

EQUATION $R = V \div I$

SUBSTITUTE $R = 6.0 \div 0.30$

ANSWER $R = 20 \Omega$

YOUR TURN

A lamp has 12 V across it and 0.40 A through it. Calculate its resistance.

KNOW $V = 12 \text{ V}$, $I = 0.40 \text{ A}$

EQUATION $R = V \div I$

SUBSTITUTE $R = 12 \div 0.40$

ANSWER $R = 30 \Omega$

WORKED EXAMPLE

A 150 Ω resistor has 3.0 V across it. Calculate the current in milliamps (mA).

EQUATION $I = V \div R$

SUBSTITUTE $I = 3.0 \div 150$

ANSWER $I = 0.020 \text{ A}$

CONVERT $0.020 \times 1000 = 20 \text{ mA}$

YOUR TURN

A 500 Ω resistor has 2.0 V across it. Calculate the current in mA.

EQUATION $I = V \div R$

SUBSTITUTE $I = 2.0 \div 500$

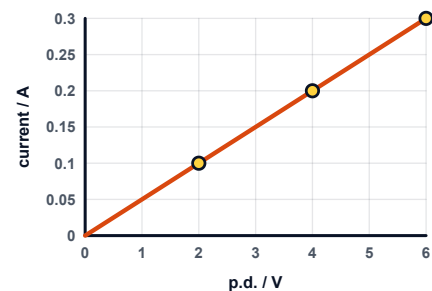
ANSWER $I = 0.0040 \text{ A}$

CONVERT 4.0 mA

OHM'S LAW: OHMIC CONDUCTORS

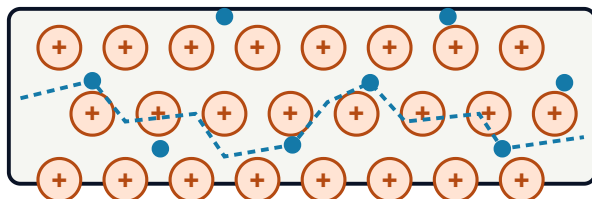
For a **fixed resistor at constant temperature**, the current is **directly proportional** to the p.d.: double the p.d. and the current doubles. The resistance stays the same.

A component like this is an **ohmic conductor**. A graph of current against p.d. is a **straight line through the origin**.



A 20 Ω resistor: straight line through the origin.

WHY IS THERE RESISTANCE?



A free electron bumps into ions as it moves: this is resistance.

Free electrons keep **colliding with the ions** in the metal. This opposes the flow of charge: resistance.

When the metal gets **hotter**, the ions vibrate more, so there are more collisions and the **resistance increases**. That is why Ohm's law needs 'at constant temperature'.



PRACTISE TRY IT

- 1 Two circuits have the same 6 V battery. Circuit A has a 10 Ω resistor; circuit B has a 30 Ω resistor. Tick **one** box.

- ☒ Circuit A has the bigger current.
☐ Both have the same current.

- ☐ Circuit B has the bigger current.
☐ You cannot tell.

- 2 **SUPPORTED** Calculate the resistance of a component with 12 V across it and 0.60 A through it.

EQUATION	$R = V \div I$	
SUBSTITUTE	$R = 12 \div 0.60$	
ANSWER	20 Ω	unit:

- 3 **SUPPORTED** A current of 0.20 A flows through a 15 Ω resistor. Calculate the p.d. across it.

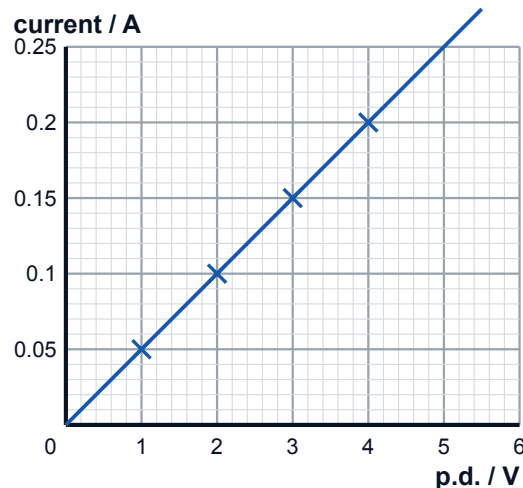
EQUATION	$V = I R$	
SUBSTITUTE	$V = 0.20 \times 15$	
ANSWER	3.0 V	unit:

- 4 **ON YOUR OWN** A 220 Ω resistor has 5.5 V across it. Calculate the current in mA.

$$I = V \div R = 5.5 \div 220 = 0.025 \text{ A} = 25 \text{ mA}$$

- 5 A student measures a component. Complete the table, plot the points and draw a line of best fit.

p.d. / V	current / A	$V \div I / \Omega$
1.0	0.05	20
2.0	0.10	20
3.0	0.15	20
4.0	0.20	20



Yes: $V \div I$ is constant (20 Ω) and the graph is a straight line through the origin, so current is directly proportional to p.d.

Is the component an ohmic conductor? Use the table and the graph.

- 6 Explain why a resistor only has a constant resistance if its temperature stays the same.

If the resistor gets hotter, the ions ...

If it heats up, the ions vibrate more, so the electrons collide with them more often and the resistance increases. So R is only constant at constant temperature.

**VIRTUAL LAB 3: DISCOVER THE RESISTANCE RULE**

Set the resistor to **10 Ω** . Change only the battery setting. Measure the p.d. across the resistor and the current.

- 7 Record and calculate.

Battery setting / V	p.d. / V	current / A	$V \div I / \Omega$
2	2.0	0.20	10
4	4.0	0.40	10
6	6.0	0.60	10
8	8.0	0.80	10
10	10	1.0	10
12	12	1.2	10

- 8 What happens to the current when the p.d. doubles? What stays constant?

The current doubles (current $\propto$ p.d.). $V \div I$ stays constant at 10 Ω , the resistance.

**EXAM CORNER**

Write in full sentences. Show every step of a calculation.

Get the marks: In part (b) use your answer to (a). Quick check: $3 \times$ the p.d. should give $3 \times$ the current if R is constant.

- 9 A fixed resistor has a p.d. of 2.0 V across it and a current of 0.10 A through it. Calculate its resistance. **[2 marks]**

EQUATION	$R = V \div I$
SUBSTITUTE	$R = 2.0 \div 0.10$
ANSWER	20 Ω unit:

- 10 The p.d. is increased to 6.0 V. The temperature stays constant. Calculate the new current. **[2 marks]**

$$I = V \div R \text{ (1)} = 6.0 \div 20 = 0.30 \text{ A (1)}$$



SUMMARY

- Resistance opposes the current. Unit: **ohm (Ω)**.
- $V = IR$** . Convert mA $\rightarrow$ A ($\div 1000$) and k $\Omega \rightarrow \Omega$ ($\times 1000$).
- Ohmic conductor: $I \propto V$ at constant temperature; the graph is a straight line through the origin.
- Hotter metal $\rightarrow$ ions vibrate more $\rightarrow$ more collisions $\rightarrow$ higher resistance.

COVER THE PAGE AND ANSWER

- | | |
|---|--|
| 1 What is the unit of resistance?
..... | 2 Write the equation linking p.d., current and resistance.
..... |
| 3 What does an ohmic conductor's I–V graph look like?
..... | 4 What happens to a metal's resistance as it heats up?
..... |

I CAN...

- | | |
|--|--|
| describe resistance and state its unit | <div><div></div><div></div><div></div></div> |
| use $V = IR$ and its rearrangements | <div><div></div><div></div><div></div></div> |
| convert mA and k Ω | <div><div></div><div></div><div></div></div> |
| decide from data or a graph whether a component is ohmic | <div><div></div><div></div><div></div></div> |
| explain why heating increases resistance | <div><div></div><div></div><div></div></div> |

KEY WORDS

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

Ammeter · Measures current. Connected in series.

Ampere (A) · Unit of current. $1\text{ A} = 1\text{ C per second}$.

Battery · Two or more cells joined together.

Charge (Q) · Carried by particles such as electrons. Unit: coulomb (C).

Conventional current · The agreed direction of current: from + to -.

Current (I) · The rate of flow of charge.

Directly proportional · If one quantity doubles, the other doubles. The graph is a straight line through the origin.

Electron · A tiny negatively charged particle. Free electrons carry the current in metals.

Ohm (Ω) · Unit of resistance.

Ohmic conductor · Current is directly proportional to p.d. at constant temperature.

Parallel · Components on separate branches. A voltmeter is connected in parallel.

Potential difference (V) · Energy transferred per unit charge between two points. Unit: volt (V).

Resistance (R) · How much a component opposes the current. Unit: ohm (Ω).

Series · Components connected one after another in a single loop.

Volt (V) · Unit of p.d. $1\text{ V} = 1\text{ J per C}$.

Voltmeter · Measures p.d. Connected in parallel, across a component.

PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

I CAN...	R · A · G
01 · Complete circuits and symbols	
explain why a circuit must be a complete loop	
recognise and draw the symbols for a cell, battery, switch, lamp, resistor and ammeter	
draw a neat circuit diagram using the rules	
connect an ammeter in series	
02 · Current and charge	
describe current as the rate of flow of charge	
calculate charge, current or time using $Q = I t$	
convert minutes to seconds before calculating	
explain why current is not used up in a loop	
03 · Potential difference	
explain p.d. as energy transferred per coulomb	
connect a voltmeter in parallel	
use $E = Q V$ and its rearrangements	
use 'p.d.s add up' in a series loop	
04 · Resistance and $V = I R$	
describe resistance and state its unit	
use $V = I R$ and its rearrangements	
convert mA and k Ω	
decide from data or a graph whether a component is ohmic	
explain why heating increases resistance	

CHECK YOURSELF

QUICK ANSWERS

Answers to the "Do now" and "Cover the page" questions, so you can check yourself when revising.
Answers to the longer questions are in your teacher's answer edition.

LESSON 1 · COMPLETE CIRCUITS AND SYMBOLS

Do now: (1) e.g. torch, phone; kettle, TV. (2) Tiny charged particles (electrons): you learn this today. (3) To break or complete the circuit, turning it off and on.

Cover the page: (1) A source (cell or battery) and a complete loop. (2) A battery is two or more cells joined together. (3) Makes a gap, so no current flows. (4) In series (in the loop).

LESSON 2 · CURRENT AND CHARGE

Do now: (1) Cell: long thin line and short thick line. Lamp: circle with a cross. (2) In series (in the loop). (3) Open switch; gap or loose wire; broken bulb; no cell.

Cover the page: (1) The rate of flow of charge. (2) Coulomb (C). (3) $Q = I t$ (4) 0.2 A

LESSON 3 · POTENTIAL DIFFERENCE

Do now: (1) The rate of flow of charge. (2) $Q = 0.50 \times 20 = 10 \text{ C}$
(3) No. The current is the same all round the loop; the lamp transfers energy.

Cover the page: (1) Energy transferred per coulomb of charge.
(2) In parallel, across the component. (3) $E = QV$ (4) 4 V

LESSON 4 · RESISTANCE AND $V = IR$

Do now: (1) 6 J of energy is transferred per coulomb. (2) Voltmeter, in parallel (across). (3) 4 V

Cover the page: (1) Ohm (Ω) (2) $V = IR$ (3) A straight line through the origin. (4) It increases.